Prediction of Road Accident Severity Using Supervised Machine Learning Techniques:
A Comparative Study on the US Accidents Dataset.

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***Abstract*—Road traffic accidents pose a serious challenge to public safety and transportation systems worldwide. Predicting accident severity in advance can support better traffic management, emergency response planning, and road safety improvements. This study proposes a supervised machine learning–based approach for predicting road accident severity using real-world environmental and road condition data. A large-scale United States accident dataset is utilized, and accident severity is formulated as a binary classification problem by categorizing severe and non-severe accidents. Key features related to weather conditions, visibility, distance, precipitation, and traffic infrastructure are selected and preprocessed to handle missing values and categorical variables. Two machine learning models which include Logistic Regression and random forest are deployed and assessed based on accuracy, precision, recall, F1-score and confusion matric. The experimental evidence indicates that the Random Forest model is more predictive than Logistic Regression especially in the prediction of severe accidents. The importance of features analysis also shows that weather and distance related factors play a role in accident severity. The findings prove that machine learning methods are effective to estimate the severity of accidents and can be used to improve road safety decision-making.**

Keywords—Road Accident Severity Prediction, Machine Learning, Random Forest, Logistic Regression, Transportation Safety, Supervised Learning

I. Introduction

In the world, road traffic accidents represent one of the major causes of injury, loss of money and death. As the number of vehicles on the roads and urbanization has increased at a very high rate, the need to know the factors that result to the severity of accidents has been a key issue of concern to transport authorities and other agencies that deal with the safety of the people. Proper forecasting of the degree of accidents would help to optimize the emergency response strategies, traffic management strategies, and develop appropriate road safety policies.

Over the last several years, the access to large-scale datasets on traffic accidents, as well as the progress in machine learning methods, has allowed researchers to study patterns in accidents more successfully. The machine learning models are especially applicable to this task since they can manipulate significant amounts of heterogeneous data and identify complex interactions between environmental, weather, and road-related variables. Machine learning techniques are usually more accurate in their predictive performance and more flexible with real-world data than conventional statistical techniques.

The study aims to predict the extent of traffic accidents through supervised machine learning models. In particular, the paper will attempt to categorize accidents by high and low severity according to the chosen environmental, weather, and road infrastructure characteristics. Two popular classification models are used, namely, Logistic Regression and Random Forest, which are implemented and compared to each other in terms of their effectiveness to predict the severity of accidents. Logistic Regression is selected because it is simple and easy to understand, and its counterpart, Random Forest, is selected because it can model nonlinear relationships and, more importantly, it can be used to interact with other features.

The data that was used in this paper was accessed on the publicly available US Accidents dataset on Kaggle. This is a curated dataset of the United States of America, which includes all traffic accidents that were recorded in 49 states between February 2016 and March 2023. It is a combination of various traffic event reporting APIs and contains millions of accident cases with their characteristics like severity level, weather, visibility, temperature, precipitation, and road infrastructure indicators. This data is very appropriate in terms of machine learning-based analysis of traffic accidents due to its large volume and the abundance of information.

To simplify the study, a subset of the relevant features was chosen from the data to concentrate on the variables that have a direct effect on the severity of accidents. Preprocessing of data, such as handling of missing values and categorical encoding, had been done to provide data quality and compatibility with the model. Severity attribute was coded into a binary target variable in which high severity was coded with the severity of 3, 4 and low severity with the severity of 1 and 2.

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The effectiveness of the applied models is measured with the help of the standard classification measures that include accuracy, precision, recall, F1-score, and confusion matrices. Also, the importance of features during the execution of the model is analyzed, and the Random Forest model is outlined to determine the most significant factors that cause accidents. The results of the conducted research are expected to shed light on the most important factors that contribute to serious traffic accidents and prove the relevance of machine learning methods in the context of traffic safety.

II. LITERATURE REVIEW

Road traffic accidents are a major issue to the health and safety status of the people of the world with millions of deaths and injuries per year. With urbanization, the number of vehicles and traffic jams growing at a high pace, the traditional methods of analyzing accidents are no longer relevant in explaining the finer details of the accidents. Consequently, the use of machine learning (ML) methods has been on the rise to forecast the occurrences, severity, and factors of an accident.

In the initial research, the main methods were based on statistics and methods of rules, like regression and probabilistic models. However, the techniques failed to capture the non-linearities between the variables related to accidents, such as weather conditions, road geometry, traffic flow and human behaviour. Recent studies have indicated that machine learning models are better predictors and have a higher adaptability when trained on large-scale accident data.

Several studies have been dedicated to predicting accidents. The Polish official records of police accidents were used in [1] to predict road accidents and collisions using a Decision Tree classifier. The findings revealed that collisions were higher than accidents and that environmental and road-related variables were also important in determining the occurrence of accidents. Even though high precision was obtained, the lack of human behavioural data was found as one of the limiting factors.

There has been a large amount of research on the prediction of the severity of accidents, and this has been especially significant in emergency response and safety planning. In [2], the UK road safety data of more than two million records were tested using the random forest, support vector machine, and neural networks. The results indicated that the highest accuracy (87-89%) was obtained with Random Forest, particularly when major and minor injuries were to be predicted. However, fatal accidents were still difficult to predict because of an imbalance in classes.

Similarly, the article in [3] also compared the severity of accidents with the use of Decision Tree, Logistic Regression, and Random Forest models. Random Forest was much better than the other classifiers, with an estimated accuracy of about 88%. The discussion has revealed that weather conditions, the state of lighting and road surface are powerful factors impacting the severity of accidents, and the issue of the impossibility of well-modelled driver behaviour was also noted.

Comparative experiments on various machine learning algorithms are becoming more and more popular. In [4], over 1.6 million accident records were used to compare the Logistic Regression, Decision Tree and Random Forest models. These findings confirmed that Random Forest gave the highest accuracy (86.86%), indicating the strength and anti-overfitting ability of this model. How hyperparameter tuning and feature selection can be used to enhance predictive accuracy was also addressed.

In addition to the accuracy of prediction, a few studies have also applied the analysis of accidents into the rescue planning and prevention systems. In [5], optimal rescue route planning was combined with accident severity prediction. The accident hotspots were determined by integrating machine learning models with clustering methods and the quickest emergency response paths were determined. This paper has shown the usefulness of ML-based accident prediction to the real-life traffic management systems.

Sustainability and environmental views have also been discussed. In [6], the analysis of road accidents through the safety and environmental perspectives was conducted with the help of the Decision Tree, LightGBM, and XGBoost models. The causes of accidents were divided into human, vehicle, road, and environmental factors, which indicated that machine learning can be used to support long-term infrastructure planning and evidence-based policy formulation.

Recent research has given special emphasis on pedestrian safety. In [7], the prediction of pedestrian crash severity was studied and the emphasis was on class imbalance which is prevalent in fatal pedestrian crash data. Comparison has been made between oversampling, class weighting and threshold-moving methods, and Random Forest with decision-level imbalance treatment gave the most dependable results. This research revealed that accuracy is not enough but recall of the severe cases is essential.

Another new direction of study has been the spatiotemporal accident analysis. Machine learning classification was used in [8] with spatial clustering algorithms like DBSCAN in identifying accident hotspots and time-based patterns. These findings indicated that the Random Forest model was the most effective in severity forecasting, whereas the clustering techniques were effective in identifying high-risk areas and times when accidents were the most prevalent.

In [9], Decision Tree, Logistic Regression, and Support Vector Classifier (SVC) were tested on a comprehensive case study of the accident prediction and prevention. The findings showed that SVC was the most accurate with a maximum of 89.83. A large amount of data visualization was also a part of the study, as well as references to several previous works, such as ensemble learning methods [10], clustering-based accident analysis [11], temporal accident research [12], and neural network-based prediction models [13].

Although machine learning methods have been widely utilized in predicting road accidents and their severity, there are various weaknesses that can be seen in the literature. The fact that many studies use country-specific datasets limits the external validity of the models proposed to work in different geographical and traffic settings. The issue of class imbalance remains a major problem especially with the proper prediction of serious and deadly accidents, since such instances are usually underrepresented in real-life data sets.

Moreover, many studies focus on the overall accuracy as the major measure of evaluation, and little consideration is paid to interpretability, feature importance, and the practical

applicability of the models. Besides that, there are relatively few studies that suggest a user-friendly, simple predictive framework that can be tested on the basis of real-world input values. These weaknesses underscore the necessity of a middle ground between strong machine learning models and interpretability and useful applicability. To fill these gaps, the current research makes use of the US Accidents data and uses the Logistic Regression and the Random Forest models on predicting the severity of accidents, focusing on analyzing the importance of features, a confusion matrix, and a practical prediction mechanism that can be applied to both scholarly research and practical implementation.

III. METHODOLOGY

A. Data Collection and Preprocessing

The US Accidents Dataset (March 2023) is the dataset that was used in the current paper and that has 7,728,394 records of traffic accidents and 46 features, such as the location of accidents, weather conditions, time, and road features. Only the most useful features were used to analyze; Severity, Distance, Temperature, Humidity, Visibility, Wind Speed, Precipitation, Weather Condition, Sunrise/Sunset, Junction, Traffic Signal, and Crossing. The severity of accidents was then translated to binary category where the severity accidents of level 3 and 4 were the high severity accidents and the low severity accidents were levels 1 and 2:

$$\text{Severity} = \begin{cases} 1 & \text{if severity level is 3 or 4 (High Severity)} \\ 0 & \text{if severity level is 1 or 2 (Low Severity)} \end{cases}$$

Missing values in the numerical characteristics were replaced with median, and categorical variables were coded through Label Encoding to change them into numeric values. The data was further divided into training and testing samples containing 80% and 20% respectively through stratified sampling to preserve the proportion of high and low severity accidents.

B. Features Selections and Model Inputs.

Following the preprocessing, the dataset was limited to 12 useful features according to their possible influence on the severity of accidents. These were the input variables to the machine learning models and Severity was taken as the target variable. To make models compatible with each other, Categorical features were encoded using label Encoding. The selection of features enabled the analysis to concentrate on the most important factors having an impact on the results of accidents without compromising the computational efficiency.

C. Model Training and Evaluation

This study used two supervised learning models, Logistic Regression and the Random Forest. The Logistic Regression equation approximates the likelihood of a severe accident based on the features of the input by the following equation:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n)}}$$

Where Y=1 indicates a high severity, X represents the input features, and $\beta_0, \beta_1, \ldots, \beta_n$ are the regression coefficients trained in the training stage.

The ensemble of decision trees is known as Random Forest, which was employed to enhance the predictive performance and minimize the overfitting. Tuning hyperparameters were used to optimize the accuracy of the model, by using the number of trees, the maximum depth, and the minimum samples per leaf. The two models were trained and tested on the training and testing sets respectively with accuracy, precision, recall, F1-score, and the confusion matrix. High-severity accidents were laid stress on the recall to demonstrate the practical significance of the proper identification of the critical cases.

D. Prediction Framework

To categorize the new accident scenarios as high or low severity, a prediction structure was created. Preprocessing of the input data based on the features selected was done to fit the format in which the model training is done. The trained Random Forest model was used to make the prediction. This framework can be applied practically to assess accident risk in the real world, which would give practical insights to road safety interventions.

E. Data Visualization

Visualisation was used to investigate data distributions, relationships between features, and model interpretation. The distribution of low and high-severity accidents was shown using count plots. The importance of the factors, as shown in the bar plots, was obtained using the Random Forest model to identify the most significant factors. Boxplots were used to analyse how the numerical variables were associated with the severity of the accidents, and confusion matrix heatmaps were used to measure the predictive accuracy of the models. These visualisations proved to be useful in the interpretation of the data as well as in the evaluation of the model that could be used to trace the trends in the severity of accidents and the contribution of particular features to predictions.

IV. RESULTS AND DETAILED ANALYSIS

A. Severity Class Distribution and Its Implications

The distribution of severity of accidents after the binary transformation indicates a strong imbalance between low severity and high severity accidents. Fig 1 demonstrates that most of the records are low severity with high-severity accidents taking up a comparatively smaller fraction of the data.

This imbalance has significant implications for model evaluation. In these cases, the overall accuracy of a model might be high because it is only necessary to predict the majority class, and even high-severity accidents still are not correctly predicted. Thus, it is not enough to use accuracy. Indicators like recall, accuracy, and the confusion table are necessary to assess the ability of the models to identify high-severity accidents which are more relevant in the analysis of road safety.

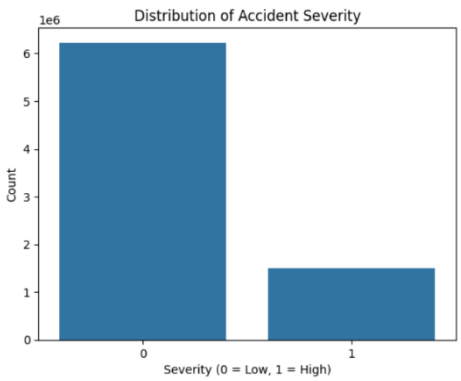

Fig. 1. Distribution of accident severity classes after binary transformation.

B. Logistic Regression Results

The Logistic Regression model had a total accuracy of about 80.53%. This, at first sight, implies predictive performance of reasonable quality. However, the classification report reveals significant limitations when undertaking deeper analysis.

The model indicate a very high recall rate for low severity accidents which implies that it has accurately categorized most of the non-severe accidents. Conversely, high-severity accidents recall was very low. This implies that the model did not identify most of the severe accidents even though it was able to identify low-risk situations correctly.

This tendency can be explained by the fact that the Logistic Regression is a linear model and can be sensitive to the imbalance between classes. The high-severity accidents are relatively rare, so the model focused on reduction of the total error instead of the studying the complex pattern related to severe outcomes. As a result, Logistic Regression is less suitable to be used when it is important to determine major accidents.

C. Random Forest Results

Random Forest model recorded a total accuracy of about 80.34% which is slightly lower than that of Logistic Regression. However, a closer analysis shows that the results of Random Forest were significantly better when it comes to high severity accidents.

The model that identifies more high-severity accidents correctly was the Random Forest model as indicated in the confusion matrix (Fig 2). This was improved in the increase of recall and F1-score of high severity.

Ensemble nature of Random Forest allows to capture non-linearity and complex interactions among features including weather conditions, distance and road infrastructure. This would allow the model to differentiate more between severe and non-severe situations involving accidents, which would make it stronger in terms of predicting the severity of the accidents in the real world.

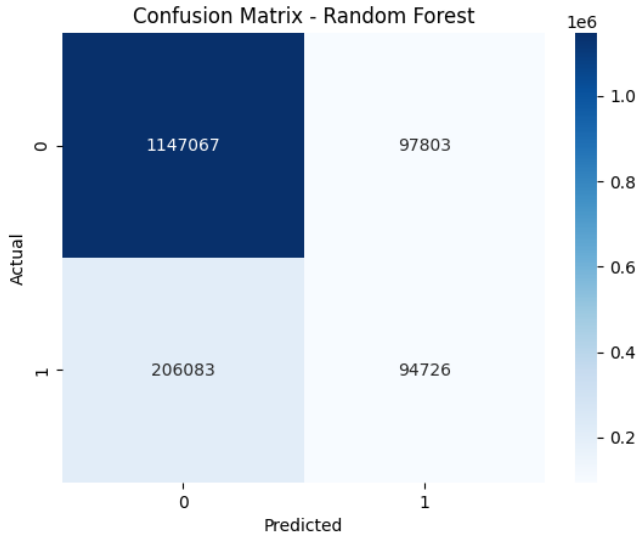

Fig. 2. Confusion matrix of the Random Forest model for accident severity classification.

D. Comparative Analysis of Models

Even though the two models yielded similar values of accuracy, their predictive behaviors varied significantly. As shown in Fig 3, accuracy is not the real effectiveness of a model in this scenario.

Logistic Regression favored the majority class and was poor at the severe accidents. Random Forest on the other hand gave a more balanced classification, with higher recall on high-severity accidents but with high-performance in low-severity accidents.

This comparison shows that the Random Forest is a better model to use in predicting the severity of accidents, especially in safety critical models where not detecting severe accidents can be costly to the user.

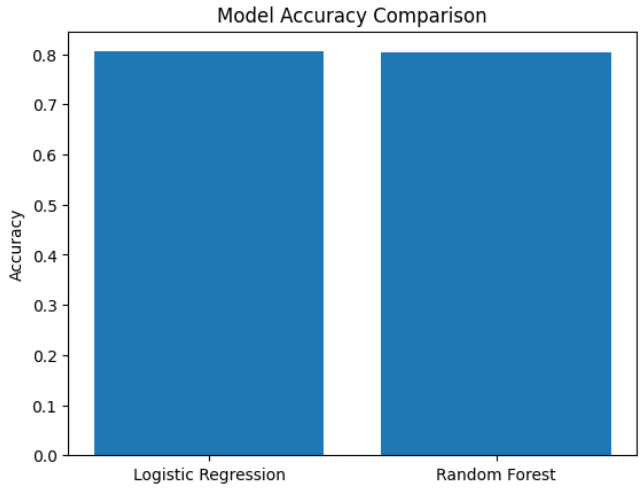

Fig. 3. Comparative performance of Logistic Regression and Random Forest models across evaluation metrics.

E. Feature Importance Interpretation

The analysis of feature importance based on the Random Forest model (Fig 4) is a helpful tool that helps to identify the factors that have a significant impact on the degree of

accidents. The most significant features were temperature, distance, and humidity, which proves that the environmental conditions are significant factors influencing the outcomes of accidents.

Wind speed and weather condition also showed importance, suggesting that poor weather conditions are a contributory factor in the severity of accidents. Other road-related characteristics like traffic signals, junctions and crossings had lower scores on importance, although their contribution does not nullify as they could be interacting with environmental factors to cause greater risk of accidents.

This discussion demonstrates the multifactorial character of the accident severity, in which chained weather-related and infrastructure-related variables contribute to the results together.

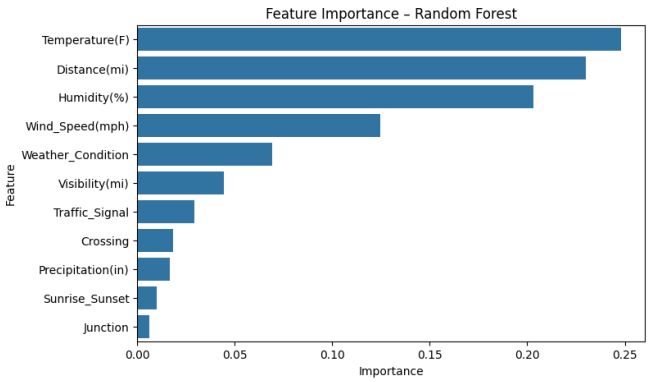

Fig. 4. Feature importance scores derived from the Random Forest model for accident severity prediction.

F. *Relationship Between Environmental Factors and Severity*

The boxplot analysis of temperature and severity of accidents (Fig 5) shows the differences in the temperature distribution between low and high severity accidents. The high-severity accidents are more likely to be found to take place over a greater range of temperatures, which implies that extreme weather conditions or changing weather conditions are likely to promote severe outcomes.

Similarly, the visual interpretation of the weather condition distributions show that some weather patterns are more commonly linked to high-severe accidents. These results substantiate the necessity of weather-related features inclusion into the predictive models based on which the severity of accidents can be assessed.

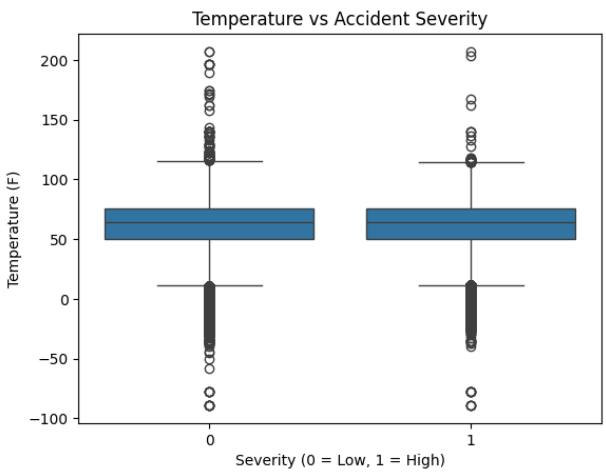

Fig. 5. Box plot illustrates the relationship between temperature and accident severity.

G. *Practical Implications of the Results*

The findings of this paper prove that machine learning models can be used to detect patterns that are linked to the severity of accidents in massive traffic datasets. The high-quality results of the Random Forest model in identifying high-severity accidents demonstrate that it can be used in practice in various ways, including traffic surveillance, early warning systems, and decision support systems related to road safety agencies.

The actionable information that this study has brought about by determining the contributing factors that include distance, humidity and temperature can also help the policymakers to develop specific interventions that can help reduce the occurrence of severe traffic accidents.

V. DISCUSSION

This paper examined the applicability of machine learning methods in predicting the degree of road accidents based on the US Accidents Dataset. The findings indicate that machine learning models are capable of effectively modeling complex relationships among environmental, road, and traffic-related variables and the intensity of accidents.

The comparative analysis reveals that the Random Forest model was more effective than the Logistic Regression especially in the detection of high-severity accidents. Even though the overall accuracy of both models was similar, Random Forest provides much higher recall and F1-score in case of severe accidents. This result is consistent with earlier research that found that the ensemble-based models like the Random Forest are better at non-linear interaction and feature dependence complexities in tasks related to the prediction of accident severity [3], [4]. The high quality of the performance of the Random Forest may be explained by its ensemble nature that decreases overfitting and makes it better than the linear models in terms of generalization.

Although the performance of Random Forest has been improved, the results also mention the inability to solve the problem of class imbalance. High-severity accidents constitute a smaller percentage of the data, which led to lesser recall of the severe cases, especially under the model of Logistic Regression. This finding align with previous studies of the severity of accidents, with such models preferring the majority group to the minority unless special imbalance-handling mechanisms are employed [2], [7]. The results of the present study also prove that the accuracy is not enough to measure the models of accident severity and other metrics, like recall and F1-score, are necessary to measure the performance on the cases of the utmost importance.

The analysis of the feature importance with the help of the Random Forest model demonstrated that such environmental and contextual factors as temperature, distance, humidity, and wind speed are important to define the level of accidents. These findings are in line with the previous studies which highlighted the importance of weather and road conditions on the outcome of accidents.[6],[8]. The significance of the traffic-related variables like traffic lights and crossings also reveal that the road infrastructure factors also lead to the probability of a serious collision, which confirms the results in previous research on traffic control and the risk of accidents [9].

The framework of prediction designed in the present study will illustrate the feasibility of machine learning models in assessing the risk of accident in the real world. The proposed framework can help traffic authorities and policymakers to identify high-risk situations and apply specific safety measures since it will allow predicting the severity of new accident scenarios. The suggestions of the similar predictive frameworks have been proposed previously as a useful tool towards enhancing road safety management, as well as emergency response planning [5].

Overall, the discussion proves that the results of this research are in line with the available literature, as well as approves the necessity of considering the imbalance in classes and prioritizing interpretability in the creation of accident severity prediction models.

VI. CONCLUSION

This study presented a machine learning-based method to predict the severity of the traffic accidents based on the US Accidents Data (March 2023). The purpose of the research was to define the patterns that relate to the high-severity traffic accidents and assess the predictive performance of various predictive models by converting the severity of the accidents to a binary classification problem and addressing the chosen environmental and road-related characteristics.

Two implemented supervised learning models, Logistic Regression and Random Forest were compared. Though the overall accuracy values of both models were close to 80%, a closer examination showed that they differ significantly in their predictive behavior. Logistic Regression was found to be good in low-severe accidents but had a low capacity of detecting high severity accidents. This weakness points to the difficulty of linear model application to real-world traffic data which is highly unbalanced and complex.

In contrast, the performance of the Random Forest model was more balanced, with higher recall and F1-score on high-severity accidents. Its ensemble form allowed the model to include non-linear relations and interactions of features thus was more useful in predicting severity. The analysis of feature importance also indicated that other environmental factors such as temperature, distance, humidity, and wind speed were important factors that affected the severity of accidents highlighting the importance of weather conditions on traffic safety.

The findings indicate that the machine learning methods and especially the ensemble-based models can be used to provide meaningful information about the severity of the traffic accidents and contribute to the data-driven decision-making. The suggested prediction framework has the potential to be used in practice in the form of the assessment of the risk of accidents, monitoring traffic, and safety planning. This study can help advance the overall goal of enhancing road safety and mitigating the effects of serious traffic accidents by conducting a study on high-risk conditions and the factors that contribute to it.

In general, this study proves that the integration of large-scale accident data with machine learning models is an effective method of learning and forecasting the severity of accidents, as well as emphasizes a need to use suitable measures of evaluation in safety-critical contexts.

VII. LIMITATIONS AND FUTURE WORK

Although good findings have been achieved in this research, several limitations are to be recognized. Firstly, the US Accidents Dataset, which is region-specific(United States) was used to analyze it. Although the dataset is huge and comprehensive, one cannot directly extrapolate the findings to other nations or other areas where the driving behavior, road infrastructure, and the traffic patterns are not the same.

Second, the problem of severity of accidents was simplified by the binary classification problem in that the severity 1 and 2 were classified as low severity and 3 and 4 as high severity. Although this method makes models more stable and less complex, it can cause the loss of detailed data about intermediate levels of severity. A multi-class severity prediction would give more specific information, but would need more sophisticated imbalance-handling methods.

Third, the data set had a high level of class imbalance as the low-severity accidents were far more than the high-severity accidents. Despite the stratified sampling approach and the application of several evaluation measures, the imbalance continued to have an impact on model performance, especially high-severity cases in the Logistic Regression model in terms of recall. No sophisticated resampling or cost-sensitive learning was used in this study.

Fourth, not all the available features were chosen to be developed into the model. Such important variables like the driver behavior, type of vehicle, the volume of traffic and real time road condition were not incorporated because of the availability of data. The lack of these variables can restrict the model to be able to capture all the underlying causes of severe accidents.

Lastly, hyperparameter optimization was minimized to ensure the simplicity of the models and their high computational performance due to the huge data size. The greater tuning or more sophisticated optimization methods might have the potential to increase model performance but was outside the bounds of the current study.

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